

MATLAB

matlab.mathworks.com

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EX1.m

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5G AND 6G LAB

LAB EXPERIMENT

New Folder

ASSEMENT QNO 1.m

EX: 2.m

ex 1.m

EX 2 .m

EX 2..m

Ex 2.m

ex 5.m

ex1.m

Workspace

Name	Value	Size	Class
d	1×200 double	1×200	double
f1	700	1×1	double
f2	3500	1×1	double
f3	28000	1×1	double
PL1	1×200 double	1×200	double
PL2	1×200 double	1×200	double
PL3	1×200 double	1×200	double

EX3.m

```
21 plot(d,PL1,'b','LineWidth',2);
22 hold on;
23 plot(d,PL2,'r','LineWidth',2);
24 plot(d,PL3,'k','LineWidth',2);
25
26 grid on;
27 xlabel('Distance (km)','FontSize',12);
28 ylabel('Propagation Loss (dB)','FontSize',12);
29 title('Spectrum Analysis of Sub-1 GHz, Mid-band & mmWave','FontSize',12);
30
31 legend('Sub-1 GHz (700 MHz)',...
32        'Mid-band (3.5 GHz)',...
33        'mmWave (28 GHz)',...
34        'Location','northwest');
35
36 %% Display Sample Values
37 disp('Propagation Loss at 1 km');
38 disp(['Sub-1 GHz (700 MHz): ',num2str(32.44+20*log10(1)+20*log10(f1)),' dB']);
39 disp(['Mid-band (3.5 GHz): ',num2str(32.44+20*log10(1)+20*log10(f2)),' dB']);
40 disp(['mmWave (28 GHz): ',num2str(32.44+20*log10(1)+20*log10(f3)),' dB']);
```

Command Window

```
Propagation Loss at 1 km
Sub-1 GHz (700 MHz): 89.342 dB
Mid-band (3.5 GHz): 103.3214 dB
mmWave (28 GHz): 121.3832 dB
>>
```

Propagation Loss Comparison

FIGURE

FORMAT

TOOLS

LAYOUT

Spectrum Analysis of Sub-1 GHz, Mid-band & mmWave

Distance (km)	Sub-1 GHz (700 MHz) [dB]	Mid-band (3.5 GHz) [dB]	mmWave (28 GHz) [dB]
0	70	85	95
2	95	110	115
4	105	118	122
6	110	120	125
8	113	122	127
10	115	123	128

36°C

Partly sunny

Search

ENG IN

14:49

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